Pitch Script

DOSM Datathon 2026 Final · Team Nexora (UTAR) · How does geopolitical risk affect tourist arrivals to Malaysia?

#	Slide	Speaker	Time	Length
1	Cover	A	0:00 – 0:25	25 s
2	Problem and objective	A	0:25 – 1:25	60 s
3	How we did it	B	1:25 – 2:15	50 s
4	Finding 1: the answer is both	B	2:15 – 3:10	55 s
5	Finding 2: forecast accuracy	B	3:10 – 3:50	40 s
6	Live Power BI demo	B	3:50 – 6:40	2 min 50 s
7	Impact	A	6:40 – 7:30	50 s
8	Conclusion and limits	A	7:30 – 8:00	30 s
9	Close	A	8:00 – 8:20	20 s
	Total		**about 8:20**	**1 min 40 s spare**

> Times assume a calm pace of about 140 words a minute. The 1 minute 40 seconds of spare time covers handovers, switching to Power BI, slow moments in the demo and nerves. Speak slowly and clearly. If you are running late, shorten the demo first, never the findings. Orange italic lines are actions, not words to say.

Slide 1 · Cover

Speaker A · 0:00 – 0:25

Good morning, judges. We are Team Nexora, four students from UTAR, with our advisor Dr. Kelwin Tan.

Our question is simple: how does geopolitical risk, like wars, conflicts and political tension, affect the number of tourists coming to Malaysia? To answer it, we used official data and machine learning on Malaysia's top 20 source markets.

Slide 2 · Problem and objective

Speaker A · 0:25 – 1:25

Tourism is very important to Malaysia. It makes up 15.9 percent of our GDP and supports 3.7 million jobs. So when risk goes up somewhere in the world, it can reach our hotels, airlines and homestays.

But here is the problem. If you look at Malaysia as one national number, the link between risk and tourist arrivals is only plus 0.08. That is almost zero. Past research, by Lee and others in 2021, also found no link at the national level. And nobody had checked each country separately.

So our objective was to test how geopolitical risk links to arrivals from each of our top 20 markets, and to forecast arrivals one to four months ahead.

Just think about this for a moment: when risk goes up, do tourists come less, come more, or both? My teammate will now explain how we found out.

Slide 3 · How we did it, in four steps

Speaker B · 1:25 – 2:15

Thank you. We did this in four steps.

Step one, we collected official open data: about 69 thousand daily rows, which we turned into 2,300 country-months.

Step two, we dated every number by the day it is actually published, not the day it happened.

Step three, we used Ridge regression to measure the link with each market, and a LightGBM model to forecast one to four months ahead.

Step four, we tested it fairly, with five rolling tests on past years and a COVID stress test, and put the results in a Power BI dashboard.

Our main rule: the model only learns from numbers that were already published, so it never sees the future.

Slide 4 · Finding 1: the answer is both

Speaker B · 2:15 – 3:10

Now our first finding. Back to the question: does risk push tourists away, or bring more in? Our answer is: both.

When we look at each country one by one, we see three groups. For 16 of our 20 markets, like Nepal, South Korea and Germany, when risk goes up, fewer tourists come. For 3 markets, Singapore, Indonesia and China, when risk goes up, more tourists come. And for Thailand, there is almost no change.

Here is our key insight. The drops and the rises cancel each other out. That is why the national link is only plus 0.08. One national number hides what is really happening, so risk must be watched market by market.

We are careful here: these are patterns, not proof that risk causes the change.

Slide 5 · Finding 2: forecast accuracy

Speaker B · 3:10 – 3:50

Our second finding is about forecasting. We tested every model on past data, as if it were live.

Our LightGBM model had the smallest error, about 16 thousand arrivals per market per month. It beat Ridge, it beat repeating last month, and it beat repeating last year.

Forecasts are most reliable one and two months ahead, with error under 12 thousand. At three and four months, the error about doubles.

And one honest result: during COVID, simply repeating last month beat our model, because no model can learn a shock it has never seen.

Slide 6 · Live Power BI demo

Speaker B · 3:50 – 6:40

Now let me show you our dashboard. It follows the four pages on the left.

Alt+Tab to Power BI. It should already be open on Malaysia Tourism Overview, full screen, both slicers set to All.

Overview (20 s). *Point at the four cards, then the Top 5 Source Countries chart.* This page gives the big picture: the latest month, the year so far, and our top five source countries. Everything here covers our 20 markets, not all of Malaysia.

1. Which markets matter? (35 s). *Click Market Analysis, then click the Non-ASEAN part of the donut.* ASEAN countries bring most of our tourists, led by Singapore, Indonesia and China. When I click Non-ASEAN, the page shows only those markets. But being big is not the same as being sensitive to risk. *Click blank space to clear.*

2. What moves arrivals? (35 s). *Click Tourism Drivers, set Source Country to Singapore, then back to All.* Here you can compare arrivals with oil prices, the exchange rate and geopolitical risk, for any market. Each chart carries a short note, because these are associations, not causes.

3. What do we forecast? (35 s). *Click Forecast and point at Forecast Arrivals 2026.* Our model forecasts about 2.28 million arrivals from our 20 markets in June 2026. It stays steady in July, then slowly drops through September. *Set source_country_name to Singapore, then back to All.* You can also see the forecast for a single market.

4. How accurate is it? (30 s). *Point at Model Accuracy by Forecast Horizon, then the COVID Stress Test chart.* The error is lowest one and two months ahead and about doubles at three and four. During COVID the simplest method won, and we show that openly.

Alt+Tab back to the slides and hand over to Speaker A. If Power BI freezes, click Demo backup and talk through the screenshots instead.

Slide 7 · Impact

Speaker A · 6:40 – 7:30

So why does this matter?

First, who could use it. Tourism Malaysia could decide which markets to promote in, and when. DOSM could add a market-level view next to the national number. And hotels, airlines and homestays could plan their staff earlier.

Second, it fits this year's theme, machine learning for sustainable tourism. It links to SDG 8.9 on tourism jobs and local culture, SDG 12.b on tools to monitor tourism, and the National Tourism Policy 2020 to 2030.

Third, it is realistic. It uses only open, official data, it updates every month with one command, and it can grow to more markets and states.

To be clear, we show how it could support decisions. We have not measured that impact yet.

Slide 8 · Conclusion and limits

Speaker A · 7:30 – 8:00

To conclude, we found three things. Geopolitical risk affects each market differently. Looking market by market shows what the national number hides. And our model forecasts best one to two months ahead.

We are also honest about our limits. These are patterns, not causes. Sudden shocks like COVID cannot be predicted. And we only cover 20 markets, so forecast ranges are our next step.

Slide 9 · Close

Speaker A · 8:00 – 8:20

So, how does geopolitical risk affect tourism in Malaysia? For 16 of our 20 markets, higher risk means fewer tourists, but not for all of them. That is why looking market by market matters.

Terima kasih. We are happy to take your questions.

Q&A answers (5 minutes)

Short answers in simple English. Backup slides: A for accuracy, B for markets, C for data.

Why do Singapore, Indonesia and China go up when risk goes up?

We don't claim risk causes it. Risk went up in the same years that tourism recovered after COVID, so we think it is mostly the recovery trend.

Is LightGBM really better than Ridge?

It had the lowest overall error, 15,969, and the lowest RMSE. But Ridge won 7 of the 20 test cells, so our choice is reasonable, not a big win.

How do you stop the model from seeing the future?

Every input is delayed by its real publishing time. The model only learns from arrivals that were already published on the forecast date.

What if the arrivals data is published later than you assumed?

We tested a 3-month and a 4-month delay. The best model stayed the same.

How important is geopolitical risk in the model?

It is the 6th most used input. Past arrivals and season matter more, which makes sense.

How would you measure SDG impact?

Our report lists indicators, such as tourism jobs, wages and local SME participation for SDG 8.9, and energy, water, waste and traffic for SDG 12.b.

Can it run without your laptop?

Yes. The Power BI file has its data built in, the data file is included, and the Python pipeline updates with one command. Code is on GitHub: github.com/william20808/nexora_dosm.